

TAP AGAINST ME

GAME DESIGN BIBLE

Every Tap Against Me Takes You Further.



Master Vision, Systems, Content, Economy, Technical Architecture, and Roadmap

Version 1.0 | July 2026

Prepared as the canonical working reference for design and development.

How to Use This Bible

This document consolidates the current design direction for Tap Against Me (TAM), including explicit decisions, accepted preferences, implementation plans, and exploratory ideas that still require testing. It is meant to prevent design drift and give development, art, audio, backend, economy, and live-operations work a shared reference.

CANONICAL Explicitly accepted decisions and non-negotiable principles. | **PLANNED** Desired features that should be built when their phase begins. | **CANDIDATE** Discussed ideas that remain subject to prototyping, balance, or scope review.

Interpretation Rule

When this document distinguishes between Competitive TAM and Infinity Taps, treat them as two distinct player experiences inside one shared universe. They may share characters, accounts, cosmetics, achievements, music identity, and progression, but their moment-to-moment design goals are different.

Document Status and Scope

- **Current product target:** Android-first, portrait-oriented, mobile-friendly 2D game made in Godot.
- **Primary experiences:** Competitive survival tapping and the no-loss Infinity Taps exploration/fidget mode.
- **Commercial model:** Free-to-play, cosmetic-first, no pay-to-win.
- **Networking direction:** Offline solo play, local play, private online matches, public matchmaking, and ranked competition with bot backfill where needed.
- **Live-operation direction:** Ninety-day seasons, seasonal passes, rotating challenges, ranked resets, and new content.
- **Technical baseline:** Godot client, Supabase identity, Cloudflare Workers/Durable Objects, and D1 persistence.

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1. Product Vision and Brand

1.1 The Game in One Sentence

Core Description

Tap Against Me is a mobile-friendly 2D tapping game where players tap to stay airborne as gravity steadily increases, collect helpful or dangerous fruit, earn Hero Coins, and compete against the environment, other players, or their own limits.

1.2 The Full Product Vision

Tap Against Me is built around an action almost anyone can understand immediately: tap the screen to keep a character moving. The simplicity is intentional. The depth comes from increasing gravity, spatial control, risk-reward pickups, survival under pressure, competition, long-term mastery, cosmetics, collectible characters, and a second mode that turns tapping into a peaceful exploratory fidget experience.

The long-term product is not just one endless survival screen. It is a shared universe in which a player can chase a personal record, climb a ranked ladder, compete with friends, collect distinctive characters and effects, display badges, explore reactive worlds, discover secrets, and return every season for new goals and cosmetics.

1.3 The Meaning of “Me”

The “Me” in Tap Against Me is symbolic rather than a single literal character.

- **In solo play**, “Me” is gravity, the player’s previous best, and the limit of their control.
- **In multiplayer**, “Me” is every opponent trying to remain airborne longer.
- **In ranked play**, “Me” is the player’s current skill ceiling and the ladder above them.
- **In Infinity Taps**, “Me” is distance, discovery, curiosity, and the player’s own willingness to keep going.

Canonical Brand Line

Me is gravity, competition, and your own limit.

1.4 Slogan and Presentation

The accepted slogan is:

Official Slogan

Every Tap Against Me Takes You Further.

- Present TAP AGAINST ME as the dominant title treatment.
- Place the slogan beneath the title in a smaller, quieter hierarchy.
- Preserve the exact wording and capitalization in English brand materials.
- A compact secondary phrase such as “Tap. Rise. Explore.” may be used for constrained placements, but it does not replace the main slogan.

1.5 Product Identity

Dimension	Identity
Platform	Android-first mobile; portrait orientation; touch-native controls.

Dimension	Identity
Genre	Arcade survival + competitive tapping + cozy exploratory fidget mode.
Session Length	Seconds to minutes for Competitive TAM; open-ended for Infinity Taps.
Tone	Playful, mischievous, cute, strange, energetic, and occasionally mysterious.
Visual Hook	Retro arcade-inspired pixel characters with expressive deformation and strong cosmetic variants.
Commercial Position	Free-to-play, cosmetic-first, no pay-to-win.
Long-Term Promise	Master the tap, collect a personal identity, compete fairly, and continually discover new worlds.

2. Design Pillars and Player Promises

2.1 Pillar: Immediate Understanding, Long-Term Mastery

A new player should understand the essential rule within seconds: tap to avoid the ground. A skilled player should still have meaningful room to improve through timing, positioning, item decisions, recovery, rhythm, and adaptation to increasing gravity.

2.2 Pillar: Every Tap Must Feel Good

The tap is not merely an input. It is the product's central tactile promise. Each tap should produce responsive motion, readable character deformation, a satisfying sound, optional haptic feedback, and a cosmetic visual reaction. Input latency or weak feedback will damage the entire game more than a missing secondary feature.

2.3 Pillar: Fair Competition

- No purchase may increase jump force, reduce gravity, improve pickup odds, grant stronger abilities, or otherwise create competitive advantage.
- Characters and skins must preserve gameplay readability and equivalent collision behavior.
- Ranked outcomes must be based on skill and valid match results, not spending.
- Bots must obey the same physics and rules as players and should never behave like perfect tapping machines.

2.4 Pillar: Two Valid Ways to Love TAM

Competitive TAM	Infinity Taps
Short, tense, outcome-driven rounds.	Relaxed, continuous, no-loss exploration.
Winning, losing, ratings, tiers, opponents.	Fidgeting, discovery, collection, world reactivity.
Standardized arenas and balance.	Large reactive spaces and playful exceptions.
Skill expression under increasing pressure.	Curiosity, movement, secrets, and persistent tap milestones.
Primary emotion: tension and triumph.	Primary emotion: comfort and wonder.

2.5 Pillar: Progress Without Pay-to-Win

Players should always have something to work toward: personal records, ranks, badges, achievements, Hero Coins, characters, cosmetics, discovery records, seasonal goals, and Infinity Taps-exclusive unlocks. Money accelerates cosmetic collection or improves advertising comfort; it does not purchase superior performance.

2.6 Player Promises

1. The game will respect rapid retries and will not place an advertisement after every loss.
2. Competitive outcomes will remain fair regardless of spending.
3. Infinity Taps rewards will retain meaning because they cannot be purchased.
4. A player's badges, skins, effects, and profile will express how they play and what they accomplished.
5. The world will respond to tapping rather than feeling like a static background.
6. New seasons will add goals and content without deleting permanently earned ownership.

3. Competitive TAM: Core Game

3.1 Core Loop

1. **Spawn and fall.** The player enters from the sky and immediately begins descending.
2. **Tap to apply upward force.** Each valid tap changes vertical momentum and may influence horizontal movement depending on the final control model.
3. **Collect or avoid pickups.** Green Apples help; Purple Poison Fruits make survival harder; Golden Hero Tokens contribute to the persistent economy.
4. **Adapt to increasing gravity.** The round becomes more difficult over time, forcing more precise timing and faster decisions.
5. **Avoid the ground.** Touching the ground ends the solo round or eliminates the player from multiplayer.
6. **Resolve and progress.** The game records survival time, match outcome, pickups, achievements, rank, Hero Coins, and personal records.

3.2 Core Physics

System	Design Direction
Gravity	Continuously or stepwise increases during the round. The curve should be legible and challenging rather than abruptly unfair.
Tap impulse	Applies upward force. It must feel immediate and consistent while allowing mastery through timing and momentum control.
Momentum	Should persist enough that tapping rhythm matters; avoid making each tap fully reset movement.
Ground collision	Ends the run or eliminates the player immediately in standard survival rules.
Top boundary	Must prevent permanent ceiling camping; use a soft push, hard boundary, or ceiling penalty after testing.
Side boundaries	May be solid, wrapping, or mode-specific. Standard ranked arenas must use one consistent rule set.

System	Design Direction
Collision shape	Equivalent across cosmetic variants, even when the art stretches, squashes, freezes, melts, or wears costumes.

3.3 Pickups

Pickup	Function	Strategic Role
Green Apple	Increases jump force.	A beneficial pickup that temporarily or cumulatively improves the player's ability to resist increasing gravity.
Purple Poison Fruit	Increases gravity acting on the player.	A deliberate hazard/debuff. It creates avoidance decisions and supports difficult achievements.
Golden Hero Token	Adds value to the persistent Hero Coin economy.	A desirable pickup that may tempt players into risky movement.

Currency Naming Standard

Use “Golden Hero Token” for the physical in-round pickup and “Hero Coins” for the wallet balance spent in the shop. One token may grant one or more Hero Coins according to balancing, but the names should remain distinct in code and UI.

3.4 Risk and Reward

Pickups should spawn in positions that create decisions rather than being automatic rewards. A Green Apple near the floor may save the player later but require a dangerous descent now. A Golden Hero Token can tempt a player away from a stable trajectory. A Poison Fruit can block a safer route or become an intentional challenge target for achievement hunters.

3.5 Skill Expression

- Tap timing and rhythm under changing gravity.
- Momentum management rather than indiscriminate rapid tapping.
- Controlled movement toward or away from pickups.
- Recovery from near-ground states.
- Risk assessment when pursuing Hero Tokens.
- Adapting after collecting one or more Poison Fruits.
- Maintaining control near boundaries and at maximum gravity.

CANDIDATE Perfect-tap windows, rhythm bonuses, wind, moving hazards, contested pickups, and gravity zones are future prototypes rather than current launch requirements.

3.6 Round Flow

Phase	Purpose
Entry	Character falls from above; player gains control immediately.

Phase	Purpose
Early stabilization	Low gravity teaches the rhythm and allows players to collect early items.
Escalation	Gravity increases and pickup decisions become more consequential.
High-pressure survival	Rapid corrections, close calls, and mastery dominate.
Resolution	Ground contact or match rule ends the round; results and rewards appear.
Fast retry	The next run should begin with minimal friction.

3.7 Standard Solo Survival

Offline solo survival is the foundational competitive-style mode. The player attempts to survive longer, improve a personal record, collect pickups, complete achievements, and learn the physics without requiring an internet connection or active opponent population.

- Primary result: survival time.
- Secondary results: fruit counts, Hero Tokens, close calls, gravity reached, achievements, and personal bests.
- Immediate replay is more important than elaborate post-round ceremony.
- Optional global and friend leaderboards may compare verified results when online.

4. Modes, Matchmaking, Bots, and Ranked Play

4.1 Launch Mode Structure

Mode	Online Requirement	Primary Use
Solo Survival	No	Practice, records, achievements, offline play.
Local Play	No or local-device dependent	Casual nearby competition and shared-device variants.
Private Online Match	Yes	Play directly with friends or invited opponents.
Casual Public Match	Yes	Quick matchmaking with bot backfill when required.
Ranked Match	Yes	Skill-based competition using Glicko-2 and seasonal tiers.

4.2 Match Formats

CANONICAL Last player remaining airborne wins the core competitive survival format. | **CANDIDATE** Additional formats may be tested after the survival foundation is stable.

Format	Description	Status
Survival	Players share a synchronized ruleset; touching the ground eliminates them; last airborne wins.	Core
Free-for-all	More than two players compete in one survival match.	Planned
Private duel	Two selected players compete directly.	Planned
Tournament/elimination	Bracket or staged competitive events.	Future
First to Zero	A specialized life/health-based ruleset from earlier exploration.	Candidate
Most Taps	A timed mode focused on valid taps or objectives rather than pure survival.	Candidate
Dual Victory	A hybrid objective format combining survival and another condition.	Candidate

4.3 Matchmaking Philosophy

A rating alone is not enough to create satisfying early matchmaking. Player survival behavior should also inform opponent selection. A player who averages roughly 15 seconds should not routinely face a player averaging 72 seconds. A player averaging 63 seconds can reasonably face a 72-second player because that matchup provides challenge without being structurally hopeless.

- Use Glicko-2 as the canonical ranked rating system.
- Track rolling survival-time performance as a secondary matchmaking signal.
- Prefer opponents with compatible rating and survival bands.
- Allow widening search ranges over time, but preserve a maximum fairness threshold.
- Use bots to prevent queues from appearing empty, especially during launch and low-population periods.

4.4 Ranked Tiers and Seasons

Tier	Suggested Divisions	Purpose
Bronze	III-I	New or developing competitors.
Silver	III-I	Players demonstrating basic control and consistency.
Gold	III-I	Competent competitors with stable survival performance.
Platinum	III-I	Strong players with advanced control.
Diamond	III-I	Highly skilled competitive players.
Master	Rating-based	Top competitive population.
Grandmaster	Leaderboard-limited	Highest seasonal recognition.

- Ranked seasons reset every 90 days.
- Core collection, owned cosmetics, lifetime achievements, and Infinity Taps progress do not reset with rank.
- Season rewards should reflect the highest rank achieved or final rank, according to the published season rule.
- Seasonal rewards remain cosmetic and may include badges, frames, skins, tap effects, particles, or titles.

4.5 Bot Backfill

Bots are a launch necessity rather than an embarrassment. They prevent cold-start matchmaking failure and allow public play even when the human population is small. They must be transparent, fair, and intentionally imperfect.

Bot Style	Behavior
Floater	Conservative, favors stable altitude and avoids unnecessary pickups.
Aggressor	Takes riskier paths toward contested items or interaction opportunities.
Climber	Prefers higher altitude and may overcorrect near the ceiling.
Trickster	Uses unusual rhythms and unpredictable movement.
Panicker	Performs well early but becomes erratic under high gravity.
Technician	Consistent and precise, but still bounded by human-like delays and mistakes.

4.6 Bot Fairness Requirements

- Bots use the same gravity, tap limits, movement physics, momentum, character stats, abilities, and arena rules as humans.
- Bots have randomized input delay and timing variance.
- Bots occasionally miss, ignore, or mishandle pickups.
- Item recognition must not be instantaneous; the current overly reactive pickup behavior should be slowed and humanized.
- Bots should be visibly identified rather than masquerading as real people.
- Beginner and casual tiers may use more bot backfill; high ranked tiers should strongly prefer human opponents.

4.7 Ghost Competition

CANDIDATE Recorded player runs can become asynchronous ghost opponents, reducing the need for simultaneous populations while preserving human movement patterns.

Ghosts are especially useful for friend challenges, leaderboard races, and regions or times with low concurrency. A ghost should reproduce validated input or trajectory data rather than simply replaying a video.

5. Infinity Taps

5.1 Purpose

Infinity Taps Promise

A no-loss, offline-friendly, relaxing mode where tapping becomes movement, exploration, fidgeting, discovery, and long-term collection rather than a test the player can fail.

Infinity Taps is not a softened version of ranked play. It is a different product fantasy. Players enter to relax, casually tap while listening to music or working, watch a lively world react, accumulate a lifetime tap total, discover places and characters, and earn items that prove they spent time in this mode.

5.2 Canonical Rules

- No traditional loss condition.
- No lives, death screen, or ground-contact failure.
- The player starts on the ground.
- Tap position influences movement left, right, upward, or downward where the environment permits.
- Lifetime and session tap totals are tracked.
- Hero Coins do not regularly spawn as pickups in this mode.
- The player earns 1 Hero Coin per 1,000 valid taps.
- Skins and effects earned in Infinity Taps cannot be purchased; they must be unlocked by playing the mode.
- The mode should function offline and synchronize appropriate progress later when accounts are available.

5.3 Movement

Tap Region	Typical Result
Left side of character/screen	Move or bounce left.
Right side	Move or bounce right.
Above	Move upward or climb.
Below / context action	Move downward, dive, descend, or interact with lower routes.
Tap and hold during Jet Engine boost	Sustained thrust rather than discrete bounces.

Movement should remain readable and playful rather than becoming a precise platformer. The character can bounce, stretch, float, dive, and react contextually. Routes may branch vertically into sky or water and horizontally into adjacent zones.

5.4 Tap Tracking

Counter	Use
Session taps	Immediate fidget progress and session summary.
Lifetime taps	Primary long-term Infinity Taps identity and milestone progression.
Biome taps	Biome mastery, journal completion, and biome-specific unlocks.
Object interaction taps	Tracks repeated reactions and secrets.
Valid taps	Inputs that count toward rewards; filters menus, accidental multitouch, and exploit patterns.

5.5 Infinity Taps Reward Philosophy

The mode should reward time and curiosity without converting into a passive coin farm. The fixed 1 Hero Coin per 1,000 valid taps provides a small universal reward, while the meaningful value comes from exclusive skins, effects, discoveries, journal records, character encounters, landmarks, and rare events.

Exclusivity Rule

Infinity Taps-exclusive cosmetics may never be placed in the normal shop, Hero Coin packs, Supporter Bundle, or Season Pass. Their value comes from being earned through this mode.

5.6 Temporary Boosts and World Modifiers

Boost / Modifier	Effect	Typical Duration
Velocity Increase	Raises travel speed through the current zone.	10–25 seconds
Jet Engine	Tap-and-hold produces sustained thrust.	10–25 seconds
Anti-Gravity	Reverses the expected tap relationship so taps force the character downward.	10–25 seconds
Extra Bouncy	Makes each tap launch the character absurdly high.	10–25 seconds
Heavy Gravity	Temporarily creates weightier, lower movement.	Candidate
Tap Burst	One tap creates several rapid movement pulses.	Candidate
Moon Bounce	Slow, floaty arcs and long hang time.	Candidate
Magnetism	Pulls nearby discoveries or interactable objects toward the player.	Candidate

6. Worlds, Biomes, Reactivity, and Discovery

6.1 Zone Structure

Infinity Taps zones should take approximately two minutes to traverse under normal movement. This duration allows a player to experience most of a biome's musical identity, environmental reactions, and landmark sequence before the next transition.

- A zone should have a readable entrance, mid-zone identity, and transition landmark.
- Routes may branch vertically or horizontally.
- Boosts can temporarily alter the expected traversal time.
- World transitions should feel continuous where possible rather than loading isolated menu-selected stages.

6.2 World Progression Examples

Route	Possible Layers
Land route	Field → forest → farmland → mountain → desert → volcanic region → ruins or fantasy region.
Ocean route	Surface → sunlight zone → twilight zone → midnight zone → abyssal zone → trench / Mariana-depth region.
Sky route	Tree level → clouds → storm clouds → upper atmosphere → near-space → orbit.
Cosmic route	Orbit → open space → Moon → nearby planets → deep-space regions → rare alien encounters.

6.3 Biome Catalog

Family	Examples
Natural	Peaceful Field, Forest, Farmland, Mountains, Desert, Volcanic Fields, Frozen Region, Autumn Valley, Rainforest Canopy, Swamplands
Subterranean / magical	Crystal Caverns, Mushroom Grove, Ancient Ruins, Dragon Highlands, Enchanted Garden, Haunted Woods
Water	Beach, Surface Ocean, Coral Reef, Twilight Ocean, Abyss, Trench
Aerial / cosmic	Cloud Kingdom, Floating Islands, Upper Atmosphere, Orbit, Moon, Planets, Deep Space
Whimsical	Candy Kingdom, Toy Room, Kitchen Counter, Library of Giants, Miniature Backyard, Inside a Giant Creature
Technological	Neon City, Arcade World, Clockwork Factory, Cyber Grid, Construction Zone, Inside a Computer
Abstract	Dreamscape, Music World, Painted World, Seasonal Festival

6.4 Living World Standard

Nearly everything visible should be considered for a tap response. The world should not merely scroll behind the character. Tapping is the player's language, so environments and creatures should answer it.

Target	Single Tap	Repeated Taps / Escalation
Rabbit	Ear twitch, glance, or small hop.	Runs away, stomps, reveals an item, leads toward a secret, or becomes annoyed.
Water	Splash and ripple.	Larger waves, bubbles, fish reaction, hidden object, or temporary current.

Target	Single Tap	Repeated Taps / Escalation
Bush	Wiggles and rustles.	Drops leaves, berries, insects, coin-like decoys, or a hidden creature.
Tree leaves	Leaves shake and fall.	Branch bends, bird flies out, fruit drops, or secret opens.
Grass	Bends or emits particles.	Creates a path, reveals tracks, or releases insects.
Rock	Small dust puff or wobble.	Cracks, exposes crystals, or reveals an underground entrance.
Hero encounter	Character reacts and becomes inspectable.	Opens purchase/unlock view if unowned; provides alternate dialogue or animation if owned.

6.5 Hero Encounters

Heroes and variants can appear naturally in the world. Tapping an unowned hero should allow the player to inspect the character and attempt to buy or unlock it when appropriate. An already owned hero should react differently—through dialogue, emotes, gifts, lore, or a variant-specific interaction—so the encounter still has value.

6.6 Discovery Journal

- Biomes and sub-biomes visited.
- Worlds and planets reached.
- Characters and variants encountered.
- Animals and repeated-tap reactions discovered.
- Landmarks and hidden routes.
- Rare events and seasonal encounters.
- Infinity Taps-exclusive cosmetics unlocked.
- Tap milestones and biome mastery.

6.7 Rare Discoveries

A 0.001% rare event concept was discussed for exceptional discoveries. Pure per-tap randomness can feel unfair and is vulnerable to automation, so the safer implementation is an encounter roll at controlled intervals—for example, every 100 valid taps—with a low base chance and a pity counter that guarantees eventual discovery.

PLANNED Rare discoveries should feel memorable and appear physically in the world. | **CANDIDATE** Exact probability, interval, and pity threshold require telemetry and fairness testing.

6.8 Example Infinity-Exclusive Rewards

Reward	Possible Requirement
Meadow Velvet Skin	Complete the field route and discover all field interactions.
Volcanic Tap Effect	Reach and master the volcanic region.
Abyssal Velvet Skin	Reach the deep trench route.
Bioluminescent Trail	Discover nocturnal or deep-ocean creatures.

Reward	Possible Requirement
Moon Dust Tap Effect	Reach the Moon and complete a lunar tap milestone.
First Contact Skin	Encounter a rare alien event.
World Traveler Badge	Visit a defined set of major biomes.

7. Characters, Velvet Hex, and Art Direction

7.1 Character Role

Characters are the emotional and collectible face of TAM. They should be instantly recognizable in motion, readable on a phone, expressive under squash-and-stretch animation, and compatible with cosmetic skins, particles, tap effects, badges, profiles, and seasonal themes.

Characters are cosmetic identities, not purchasable power. Their gameplay footprint and competitive behavior must remain equivalent unless a future ability system is introduced equally and balanced independently from monetization.

7.2 Velvet Hex

Character Lore

Velvet Hex — A stitched tuxedo fiend that trades cuddles for tiny curses.

- Core silhouette: tuxedo-cat body, plush/slime-like triangular mass, cat ears, two demon horns, pointed tail, button eyes, white tuxedo chest, and purple bow tie.
- Core personality: cute, mischievous, strange, cuddly, and faintly dangerous.
- Animation identity: squash, stretch, wobble, float, and splat-like landing while remaining recognizably Velvet Hex.
- The body can be interpreted as plush, slime, ice, toxic ooze, royal fabric, arcade material, or costume without losing the core silhouette.

7.3 Art Direction

CANONICAL Retro 1990s arcade-inspired pixel art with compact sprite resolution and visible pixel clusters. |

PLANNED All variants should preserve a consistent silhouette, frame layout, and collision footprint.

- Avoid high-resolution illustration disguised with a pixel filter.
- Use crisp nearest-neighbor scaling and no texture filtering.
- Avoid anti-aliased edges and painterly gradients.
- Use limited, theme-driven palettes with strong silhouettes and readable highlights.
- Design at a small native sprite size and scale by integer multiples in-game.
- Keep the eyes, bow tie, white tuxedo chest, ears, horns, and pointed tail recognizable across variants unless the theme deliberately transforms one element.
- Do not bake checkerboard transparency into final production sprites; export true alpha.

7.4 Standard Jump Sprite Sheet

Frame	Pose	Purpose
1	Neutral / ready	Baseline idle or airborne-ready silhouette.
2	Compression / squash	Anticipation before upward force.

Frame	Pose	Purpose
3	Launch / stretch	Character elongates as movement begins.
4	Rising / airborne	Body regains shape while ascending.
5	Apex / suspended	Readable mid-air frame before descent.
6	Landing / impact	Squash, particles, or theme-specific impact reaction.

Production sheets should use identical frame dimensions, baselines, pivot points, and ordering across skins so the game can swap a variant texture without changing animation code.

7.5 Velvet Hex Variant Catalog

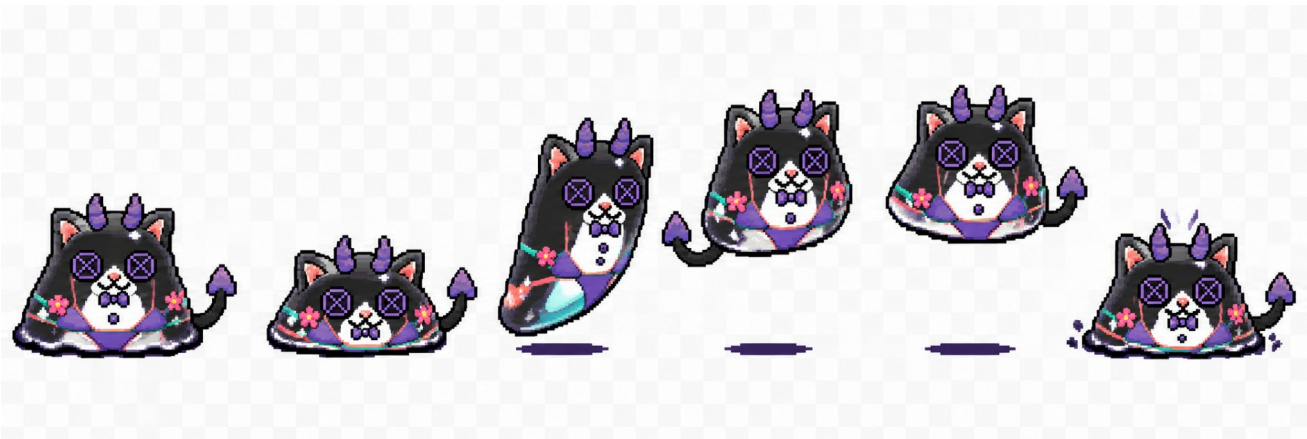
Variant	Theme Requirements
Classic Velvet Hex	Black-and-white tuxedo slime/plush demon; purple horns, button eyes, purple bow tie.
Scarlet / Vampire Hex	Dracula-inspired red, black, and purple-trimmed trench coat; red horns; visible vampire teeth; tuxedo chest and tie retained.
Beach Hex	Beachwear/bikini-inspired theme with bright tropical accents while retaining tuxedo chest and bow tie.
Toxic Hex	Neon toxic green ooze, bubbles, drips, green ear highlights, contaminated horns, and toxic splash landing.
Golden Royalty / Royal Hex	Gold and purple royal trim, crown/horn ornamentation, regal cloak accents, and gold-edged silhouette.
Arcade Hex	Neon cyan, magenta, and purple details, arcade icons, pixel sparks, and retro machine energy.
Frozen Hex	Deep ice coverage, ice-shard horns, ice-cube eyes, frosty edge buildup, blue bow tie, snow and ice particles.
Future candidates	Molten, Ghost, Corrupted, Strawberry Slime, and other seasonal or biome-specific variants.



Classic Velvet Hex — six-frame jump progression and the silhouette standard for variants.



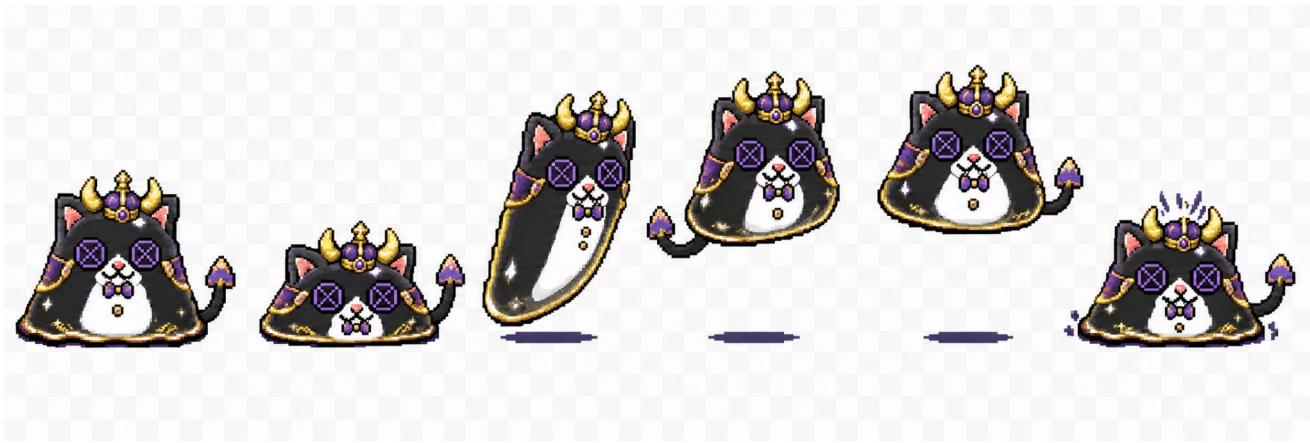
Scarlet / Vampire Hex — Dracula-inspired costume, red horns, coat, fangs, tuxedo chest, and bow tie.



Beach Hex — tropical/beachwear variation while retaining the recognizable core identity.



Toxic Hex — contaminated green ooze, bubbles, drips, and toxic impact treatment.



Golden Royalty / Royal Hex — gold and purple crown, trim, and regal styling.



Arcade Hex — neon arcade iconography and cyan-magenta highlights.

7.6 Frozen Hex Production Correction

Frozen Hex must use the same six-frame sheet structure and clean light/transparent presentation as the approved Classic, Scarlet, Beach, Toxic, Royal, and Arcade sheets. The theme requires deep ice coverage, ice-shard horns, ice-cube eyes, frosted edges, ice particles, and a readable tuxedo chest/bow tie. Atmospheric dark backgrounds or incomplete frame counts are concept-art errors, not the production standard.

8. Cosmetics, Tap Effects, and Particle Effects

8.1 Cosmetic Taxonomy

Category	Examples
Character skins	Classic, Toxic, Royal, Arcade, Frozen, Scarlet, Beach, seasonal and biome variants.
Character particle effects	Trails and particles attached to movement, idle, jump, and landing.
Tap effects	Visual reaction at the screen location where the player taps.
Spawn / entrance effects	How a character enters a match or world.
Landing / elimination effects	Impact, defeat, or round-resolution presentation.

Category	Examples
Profile cosmetics	Badges, frames, titles, banners, and username treatments.
Victory presentation	Poses, emotes, and result-screen animations.
Environment cosmetics	Background themes, fruit reskins, and arena decorations where readability remains intact.

8.2 Canonical Tap Effects and Prices

Effect	Behavior	Price
Purple Pulsing Rings	Rings pulse outward from each tap. Default purple version.	100 Hero Coins
Additional Ring Color	Unlocks another selected color; color picker supports brightness and RGB selection.	50 Hero Coins
RGB Pulsing Rings	Ring color changes randomly on each pulse/tap.	300 Hero Coins
Rainstorm	Rain falls at the tap location; a 0.001% lightning strike may burn the ground.	200 Hero Coins
Snowfall	Snow falls at the tap location and can visually accumulate on the ground.	200 Hero Coins
Magma / Lava	Lava or magma reaction spreads from the tap.	200 Hero Coins
Fire	A compact flame burst at the tap location.	75 Hero Coins
Autumn Leaves	Leaves scatter or spiral from the tap.	200 Hero Coins

8.3 Candidate Tap Effects

The following effects were discussed as future cosmetic content. Their prices and final behavior are not yet canonical:

- Bubble Pop, Pixel Lightning, Starfall, Galaxy Portal, Ghost Hands, Slime Splash, Ink Blot, Paint Burst.
- Flower Bloom, Mushroom Growth, Crystal Growth, Mini Earthquake, Water Splash, Ocean Wave.
- Biome-specific and seasonal effects tied to Infinity Taps, achievements, or passes.

8.4 Character Particle Effects and Prices

Particle Effect	Price
Purple Slime Droplets	100
Shadow Smoke	150
Golden Sparkles	150
Poison Bubbles	125

Particle Effect	Price
Pixel Stars	100
Fire Trail	175
Blue Ghost Fire	200
Snowflake Trail	150
Autumn Leaf Trail	150
Bubble Trail	125
Rainbow Trail	300
RGB Pixel Trail	300

A player equips one primary character particle effect at a time. Theme-specific skins may include a built-in subtle material effect, but the paid/equipped particle slot should remain separately visible and configurable.

8.5 Cosmetic Readability and Performance

- Effects must not obscure the character, ground, pickups, or opponent trajectories.
- Provide reduced-particle and effects-off accessibility options.
- Competitive modes may clamp effect density while preserving ownership and style.
- Particle systems should pool objects rather than allocate continuously during rapid tapping.
- Tap sounds, particle sounds, and cosmetic sounds should be independently adjustable.

8.6 Acquisition Sources Must Remain Distinct

Source	Rule
Normal shop	May be bought with earned or purchased Hero Coins.
Achievements	Earned only by completing the associated accomplishment.
Infinity Taps	Earned only by playing and discovering within Infinity Taps; never sold.
Season	Available only during the original season; ownership remains permanent once earned.
Supporter bundle	Permanent bundle cosmetics tied to the specific bundle.
Ranked rewards	Earned through seasonal competitive performance.

9. Progression, Economy, and Shop

9.1 Progression Tracks

Track	Examples
Performance	Best survival time, maximum gravity reached, close calls, match wins, ranked rating.
Activity	Rounds played, taps, lifetime Infinity Taps total, daily streaks.
Collection	Characters, variants, skins, tap effects, particles, badges, profile items.
Exploration	Biomes, worlds, landmarks, animals, rare events, journal completion.
Economy	Hero Coins earned, held, and spent; shop purchases.
Recognition	Achievements, badges, ranked tiers, seasonal rewards, completion percentages.

9.2 Hero Coin Sources

- Golden Hero Tokens collected during competitive gameplay.
- One Hero Coin per 1,000 valid taps in Infinity Taps.
- Achievements and challenge rewards.
- Season pass and free seasonal track rewards.
- Optional rewarded advertising where permitted by final ad design.
- Direct Hero Coin purchases.

The competitive pickup-to-coin conversion and average daily earn rate remain balancing variables. Infinity Taps has the one-per-1,000-taps canonical rate.

9.3 Direct Hero Coin Pricing

Package	Price	Hero Coins	Role
One-time Starter Pack	\$0.99	200	First-purchase value; one per account.
Hero Coin Pocket	\$0.99	100	Repeatable base-rate package.
Hero Coin Pouch	\$4.99	600	Small collector package.
Hero Coin Chest	\$9.99	1,300	Mid-tier package.
Hero Coin Vault	\$24.99	3,500	Large themed-collection package.
Hero Treasury	\$99.99	15,000	Long-term collector/supporter package.

9.4 Cosmetic Price Bands

Item	Suggested Hero Coin Range
Basic color variation	50–100
Basic tap effect	100–200
Premium tap effect	250–400
Basic character particle effect	200–350
Premium particle effect	400–600
Standard character skin	400–700
Premium animated skin	800–1,200
Character cosmetic bundle	1,200–2,000
Premium themed collection	2,000–3,500

9.5 Monetization Products

Product	Price	Contents / Rule
30-Day Ad-Free Pass	\$2.99	Removes forced/non-optional ads for 30 days; not permanent.
Season Pass	\$4.99	Premium seasonal reward track with cosmetics obtainable only during that season.
Supporter Bundle	\$9.99	Character bundle, character particle effect, tap effect, and 60 days of ad-free access.
Character skins	Variable	Cosmetic only.
Particle and tap effects	Variable	Cosmetic only.

9.6 No Pay-to-Win Policy

Non-Negotiable

Players may spend money on cosmetics, presentation, currency, passes, and temporary ad removal. They may not buy stronger physics, lower gravity, higher jump force, better item odds, ranked advantages, exclusive competitive abilities, or superior characters.

9.7 Advertising Guardrails

- Never interrupt an active round with an advertisement.

- Do not show a forced ad after every loss; frequent deaths make this especially harmful to TAM's rapid-retry loop.
- A practical starting cap is one forced ad after several solo rounds with a real-time cooldown, subject to retention testing.
- Players with an active ad-free entitlement see no forced ads.
- Optional rewarded ads may remain available for clearly disclosed rewards, provided they do not create ranked advantage.

9.8 Shop Experience

- Categories: Featured, Characters, Skins, Tap Effects, Particles, Profile, Hero Coins.
- Every item shows preview, compatibility, price, ownership, rarity, source, and seasonal status.
- Tap effects should have an interactive tap-preview area.
- Particle effects should preview the character jumping and moving.
- Purchase confirmation must distinguish direct-money products from Hero Coin purchases.
- Restore purchases and entitlement expiration must be supported on reinstall and device change.

Achievements, Badges, and Challenges

10.1 Purpose

Achievements create long-term goals beyond a single score or match. They reward experimentation, mastery, persistence, collection, unusual decisions, and participation in both Competitive TAM and Infinity Taps. Rewards may include Hero Coins, badges, skins, tap effects, particle effects, or other cosmetics.

10.2 Badge Use

- A badge is a permanent profile collectible.
- Players equip one primary badge beside their username.
- Players display up to three additional showcase badges on the full profile.
- Primary badges may appear in multiplayer introductions, results, and leaderboards where space permits.
- Badges are not consumed, can be changed at any time, and must show their unlock source.
- Tiered badges upgrade through Bronze, Silver, Gold, and Mythic versions; the highest version is the default display.

10.3 Achievement Catalog

Achievement	Requirement	Reward
Life of Poisonous Fruit	Consume 500 Poison Fruits across all matches.	10 Coins + Poison Fruit Badge
Anti-Gravity	Consume 500 Green Apples.	10 Coins + Golden Apple Badge
Fruit Salad	Collect a Green Apple and Poison Fruit within five seconds.	5 Coins + Fruit Salad Badge
Toxic Appetite	Consume five Poison Fruits during one round.	10 Coins + Toxic Slime Badge
Poison-Proof	Survive 30 seconds after consuming three Poison Fruits in one round.	15 Coins + Poison Shield Badge
Apple a Day	Collect at least one Green Apple on seven consecutive days.	15 Coins + Apple Calendar Badge
Orchard Raider	Collect 25 Green Apples during one round.	20 Coins + Apple Basket Badge
Clean Eating	Survive two minutes without touching a Poison Fruit.	10 Coins + Pure Apple Badge

Achievement	Requirement	Reward
No Boost Needed	Survive 90 seconds without collecting a Green Apple.	15 Coins + Natural Flyer Badge
Bad Decisions	Consume three Poison Fruits before the first Green Apple.	5 Coins + Questionable Fruit Badge
First Flight	Survive 30 seconds.	5 Coins + Tiny Wings Badge
Learning to Fly	Survive one minute.	10 Coins + Silver Wings Badge
Gravity Defier	Survive two minutes.	20 Coins + Gravity Badge
Sky Eternal	Survive five minutes.	50 Coins + Eternal Wings Badge
Terminal Velocity	Survive 20 seconds after gravity reaches maximum.	25 Coins + Falling Star Badge
Close Call	Recover after entering the configured close-call ground distance.	5 Coins + Scraped Paw Badge
Living on the Edge	Trigger five close calls during one round.	15 Coins + Razor Edge Badge
Perfect Control	Survive one minute without touching top or side boundaries.	15 Coins + Balance Badge
Untouchable	Survive one minute without collecting any item.	15 Coins + Empty Sky Badge
One More Tap	Remain near the ground for ten continuous seconds and survive.	10 Coins + Last Tap Badge
Coin Collector	Hold 500 Hero Coins at one time.	25 Coins + Coin Collector Badge
Golden Hoard	Hold 2,500 Hero Coins at one time.	50 Coins + Golden Vault Badge
Token Hunter	Collect 1,000 Hero Coins through gameplay.	25 Coins + Token Hunter Badge
Big Spender	Spend 500 Hero Coins in the shop.	15 Coins + Shopping Bag Badge
Shop Regular	Purchase ten different shop items.	20 Coins + Merchant Badge
Fully Dressed	Equip a skin, trail, badge, and tap effect simultaneously.	10 Coins + Wardrobe Badge
Golden Opportunity	Collect three Golden Hero Tokens during one round.	15 Coins + Golden Hero Badge
Shiny Distraction	Lose within two seconds of collecting a Golden Hero Token.	5 Coins + Dropped Coin Badge
First Victory	Win the first online match.	10 Coins + First Crown Badge
Last One Floating	Win ten online matches.	25 Coins + Survivor Crown Badge
Sky Champion	Win 100 online matches.	100 Coins + Champion Wings Badge
Photo Finish	Win after the final opponent is eliminated within one second of the player touching ground.	20 Coins + Photo Finish Badge

Achievement	Requirement	Reward
Flawless Victory	Win without consuming a Poison Fruit.	15 Coins + Perfect Crown Badge
Poisoned Champion	Win after consuming at least three Poison Fruits.	25 Coins + Toxic Crown Badge
Comeback Flight	Become the lowest active player and still win.	15 Coins + Comeback Badge
Crowded Skies	Win a match with the maximum supported player count.	25 Coins + Crowded Sky Badge
Rivalry	Defeat the same player in three completed matches.	10 Coins + Rival Badge
Friendly Competition	Complete ten private matches with friends.	15 Coins + Friends Badge
Getting Started	Complete five achievements.	10 Coins + Beginner Badge
Badge Collector	Unlock ten badges.	20 Coins + Badge Case Badge
Achievement Hunter	Complete 25 achievements.	50 Coins + Achievement Hunter Badge
Completionist	Complete every enabled standard non-seasonal achievement.	Exclusive Skin + Completionist Badge
Dedicated Tapper	Play 100 completed rounds.	20 Coins + Tap Counter Badge
Tap Veteran	Play 1,000 completed rounds.	75 Coins + Veteran Badge
Persistent	Immediately restart after losing 25 separate rounds.	15 Coins + Never Quit Badge
Daily Flyer	Play at least one completed round on seven consecutive days.	20 Coins + Weekly Wings Badge
Was That Worth It?	Secret: lose while actively moving toward a Poison Fruit.	5 Coins + Regret Badge
Floor Inspector	Secret: lose within three seconds of starting.	Floor Badge
Maximum Poison	Secret: consume seven Poison Fruits in one round.	25 Coins + Purple Skull Badge
Perfectly Balanced	Secret: end with equal Green Apple and Poison Fruit counts.	10 Coins + Balance Fruit Badge
Velvet's Nine Lives	Secret: trigger nine close calls in one round while using Velvet Hex.	25 Coins + Nine Lives Badge
Too Greedy	Secret: lose within one second of collecting a Golden Hero Token.	Greedy Hero Badge

10.4 Tiered Achievement Example

Tier	Poison Fruits Consumed	Reward
Bronze	25	3 Coins

Tier	Poison Fruits Consumed	Reward
Silver	100	5 Coins
Gold	500	10 Coins + Poison Fruit Badge
Mythic	2,500	50 Coins + Animated Poison Badge

10.5 Daily and Weekly Challenges

Daily Examples	Weekly Examples
Collect 10 Green Apples.	Survive for a combined 20 minutes.
Survive for 60 seconds.	Win 10 online matches.
Complete five rounds.	Collect 100 Green Apples.
Collect three Golden Hero Tokens.	Consume 25 Poison Fruits.
Trigger two close calls.	Earn 100 gameplay Hero Coins.
Finish a round without collecting fruit.	Complete five daily challenges.
Consume three Poison Fruits.	Play 50 rounds.
Play one online match.	Trigger 30 close calls.

Suggested reward bands are 2–5 Hero Coins per daily challenge and 10–25 Hero Coins or equivalent value per weekly challenge. Exact values should be balanced against actual earn rates and cosmetic prices.

10.6 Achievement Architecture

Autoload / Service	Responsibility
EventBus.gd	Receives gameplay signals without embedding achievement logic in gameplay objects.
PlayerStats.gd	Maintains lifetime, round, and streak statistics.
AchievementManager.gd	Evaluates definitions, progress, completion, and rewards.
ChallengeManager.gd	Rotating daily/weekly definitions, expiration, and progress.
BadgeManager.gd	Ownership, primary badge, and three showcase slots.
ProfileManager.gd	Profile presentation and cosmetic loadout.
SaveService.gd	Local persistence and later cloud synchronization.

- Gameplay emits events such as round_started, round_ended, item_collected, close_call_triggered, coin_balance_changed, shop_item_purchased, and online_match_completed.

- Achievement definitions are data-driven and include ID, name, category, tracked stat, target, comparison, hidden flag, tier, rewards, retroactive flag, and version.
- Rewards must be idempotent. A unique transaction key prevents duplicate Hero Coin grants after reload or reconnect.
- Coin-balance achievements must evaluate the pre-reward balance so the achievement reward itself cannot cause completion.
- Online and ranked achievements require server-confirmed match results.

11. Seasons and Live Operations

11.1 Season Cadence

TAM seasons last 90 days. Ranked progression resets on season transition, while permanent collections, achievements, owned cosmetics, lifetime statistics, and Infinity Taps progression remain.

11.2 Seasonal Pass

Track	Availability	Content
Free track	All players	Hero Coins, standard seasonal rewards, one or more accessible cosmetics, challenge rewards.
Premium track	\$4.99 Season Pass	Unique skins, particles, tap effects, animated badges, frames, and additional Hero Coins.

- Purchasing the pass unlocks the premium track; it does not instantly grant every reward.
- Players progress through gameplay and seasonal challenges.
- Season cosmetics are only obtainable during the original season.
- Unlocked items remain permanently owned after the season ends.
- Unfinished rewards become unavailable when the season closes.
- Season-exclusive content should not later enter the standard Hero Coin shop.

11.3 Seasonal Content Types

- Character skins and variants.
- Character particle effects and tap effects.
- Profile frames, badges, titles, and banners.
- Victory or entrance animations.
- Infinity Taps seasonal worlds and rare encounters.
- Ranked seasonal badge and end-of-season rewards.
- New biome, creature, character, or mechanic where scope permits.

11.4 Live-Operations Guardrails

- Do not design seasons around mandatory daily attendance.
- Clearly display season end time and progress requirements.
- Use UTC for server challenge and season timing.
- Do not remove already earned ownership when a season ends.
- Avoid flooding the game with so many limited items that individual cosmetics lose identity.

12. Music, Audio, Haptics, and Performance

12.1 Audio Identity

TAM's audio should combine playful electronic energy, bouncy rhythm, tactile tapping, and small organic details. The same musical DNA can appear across menus, competition, and Infinity Taps while each mode uses a different intensity and arrangement.

12.2 Infinity Taps Main Loop Music

Accepted Direction

Playful, energetic, slightly mischievous, and endlessly replayable: bright electronic percussion, bouncy synth bass, plucked marimba or kalimba melodies, soft arcade tones, and small organic sounds with occasional unexpected reactions.

12.3 Main Menu Music

Accepted Direction

A calmer and cleaner introduction to the same musical identity: upbeat, welcoming, lightly mischievous, with soft electronic percussion, warm bouncy synth bass, gentle marimba or pizzicato melodies, subtle arcade tones, and organic accents.

12.4 Solo and Competitive Music

Accepted Direction

Faster, sharper, and more focused than Infinity Taps: punchy electronic percussion, driving synth bass, quick plucked melodies, soft arcade tones, and short organic accents that support tension without becoming exhausting.

12.5 Biome Music

Infinity Taps zones are approximately two minutes long so players can experience most of a song or layered arrangement. The preferred system uses a shared rhythmic/bass identity with biome-specific melody, ambience, instrumentation, and environmental stems. This preserves cohesion and reduces file duplication.

Biome	Possible Audio Layer
Forest	Wood percussion, leaves, birds, warm marimba.
Ocean	Filtered synths, bubbles, water ambience, softer attack.
Frozen	Bell-like tones, crystalline plucks, wind, ice cracks.
Volcanic	Low percussion, rumble, distorted bass, ember accents.
Arcade	Chiptune accents, coin-like tones, neon synth layers.
Space	Sparse percussion, wide pads, radio-like organic details.

12.6 Sound Effects and Haptics

- Each tap may trigger a soft core sound plus cosmetic effect audio.
- Green Apples, Poison Fruits, Hero Tokens, close calls, maximum gravity, achievements, and discoveries require distinct cues.
- Infinity Taps environmental objects should have layered single-tap and repeated-tap sounds.
- Haptics should be subtle, optional, and scaled by event importance.
- Separate settings: music, ambience, tap sounds, character sounds, cosmetic effect sounds, and haptics.

12.7 Music Compression and Import

Requirement	Direction
Masters	Preserve WAV or FLAC source masters.
Shipping codec	Ogg Opus, stereo, 48 kHz, variable bitrate.
Calm tracks	Approximately 80–96 kbps.
Competitive tracks	Approximately 96–112 kbps.
Import behavior	Stream long music or use compressed-in-memory; do not decompress all tracks on load.
Avoid	Recompressing MP3, unnecessarily low sample rates, and forced mono conversion without listening tests.

The current rough soundtrack discussed was about eight songs and approximately 27 MB. A practical shipped target is 10–15 MB using Opus compression. A more aggressive 5–8 MB target is possible if the music system reuses shared stems and shorter biome layers rather than storing fully independent tracks.

13. User Experience, Interface, and Accessibility

13.1 Mobile Presentation

- Portrait orientation is the primary design target.
- Touch targets must be comfortable for one-handed play.
- The active playfield should remain visually dominant.
- Menus should not require precision tapping or dense desktop-style layouts.
- Use anchors and containers so UI adapts to varying phone aspect ratios and safe areas.

13.2 Primary Screens

Screen	Key Functions
Main Menu	Competitive play, Infinity Taps, character, shop, achievements, profile, settings.
Mode Select	Solo, local, private, casual, ranked, Infinity Taps.
Character / Loadout	Character, skin, particle, tap effect, badge, profile cosmetics.
Shop	Preview and purchase cosmetic content or Hero Coins.

Screen	Key Functions
Achievements	Progress, filters, rewards, hidden achievements, completion totals.
Profile	Primary badge, three showcase badges, statistics, rank, cosmetics, discovery highlights.
Infinity Journal	Biomes, characters, creatures, landmarks, events, and exclusive unlocks.
Post-Round Results	Survival, pickups, match result, rank change, achievements, retry.

13.3 Achievement and Badge UX

- Show name, description, current progress, target, reward, category, and completion state.
- Automatically grant rewards rather than requiring manual claims that can be forgotten.
- During gameplay, only show compact, non-obstructive progress notices.
- Show full achievement completion after the round or in a small banner.
- Secret achievements conceal their requirement until completion.

13.4 Accessibility

- Reduced particle density and option to disable cosmetic effects.
- Independent sound and haptic controls.
- Color-independent pickup shapes and icons so benefits/hazards are not communicated by color alone.
- Readable font size and contrast on small screens.
- Avoid excessive screen shake or flashing; provide reduced-motion settings.
- Support pause and offline continuity where the mode allows it.

13.5 Pixel Display Settings

- Use a fixed low-resolution game canvas appropriate for portrait mobile and integer scaling where possible.
- Import character and icon textures with nearest-neighbor filtering.
- Disable mipmaps for small pixel sprites unless a specific zoom use requires them.
- Keep sprite pivots and collision shapes consistent across variants.

14. Technical Architecture and Current Backend State

14.1 Client

Component	Direction
Engine	Godot.
Primary platform	Android-first; portrait mode.
Input	Touch; mouse input used for desktop development; optional controller support may be considered later.
Local save	Versioned player profile, statistics, achievements, challenge progress, and Infinity Taps progress.

Component	Direction
Networking	Godot client communicates with Cloudflare Worker services and Supabase identity.

14.2 Backend Stack

Service	Role
Supabase	Account creation, email authentication, Google OAuth, and signed identity tokens.
Cloudflare Workers	Public API, validation, profile operations, rewards, and service orchestration.
Durable Objects: LOBBY	Matchmaking queue and lobby coordination.
Durable Objects: MATCH_ROOM	Synchronized online match room state.
Cloudflare D1 binding DB	Persistent users, profiles, achievements, badges, rewards, matches, flags, and related records.

14.3 Current Production State

Item	Current State
Repository	tap-against-me
Worker URL	https://tap-against-me.dontwatchmedia.workers.dev
D1 database name	tap-against-me
D1 database ID	3d28400f-6c8c-4fd3-837e-390f4defd7d6
Bindings	MATCH_ROOM, LOBBY, DB
Lobby wait	LOBBY_WAIT_MS = 8000
Countdown	COUNTDOWN_MS = 3000
Automated tests	53 passed, 0 failed after Windows integration-test fix.
Schema route	Used for initialization, then disabled/returned 404 in production.
Health	Production health endpoint confirmed operational.

14.4 Authentication Direction

The backend moved from an earlier HS256 secret-verification approach to Supabase JWKS/public-key verification using ES256. The production implementation should reject invalid or unverified identity and use the token subject as the authoritative account identity.

- Godot configuration needs the public Supabase URL/publishable key and Worker base URL.
- Test real email signup, confirmation, login, token refresh, and D1 profile provisioning.
- Configure and validate Google OAuth on Android.
- Test two phones with separate accounts entering the same online flow.

14.5 Multiplayer Authority

- The server should validate match configuration, participants, timing, and final results.
- The client can simulate responsive local motion, but ranked outcomes must be server-confirmed.
- Durable Objects should coordinate room state, countdown, disconnect handling, and result finalization.
- Bots should run from deterministic or server-validated logic using the same simulation rules.

14.6 Suggested Godot Module Structure

Module	Role
game_config.gd / match_config.gd	Central constants, mode rules, and configurable thresholds.
player.gd	Input, movement, animation state, and collision response.
pickup.gd	Pickup behavior and event emission.
player_stats.gd	Round/lifetime stats.
bot_brain.gd	Bot decision and input generation.
net_service.gd / net_match.gd / online_match.gd	Online service and synchronized match flow.
challenge_manager.gd	Rotating challenge progress.
save_service.gd	Local versioned persistence.
ui.gd / stats_panel.gd	HUD, results, stats, and menus.

14.7 Android Device Testing Workflow

1. Install Godot export templates, Android SDK, and OpenJDK.
2. Configure the Android SDK and Java paths in Godot.
3. Enable USB debugging on the Android phone and authorize the development computer.
4. Create an Android export preset with a unique package identifier and mark it runnable.
5. Use one-click deploy to export, install, and launch the debug build.
6. Use remote debugging to inspect output, errors, networking, touch input, scaling, and performance on the actual phone.

15. Data, Security, Anti-Cheat, and Analytics

15.1 Server-Authoritative Assets and Rewards

- Currency purchases, reward transactions, badge ownership, and online match results should be stored and validated server-side.
- A loadout must be sanitized so a user can equip only skins, effects, and badges the server knows they own or earned.
- Achievement reward transactions use unique IDs to prevent duplicate grants.
- Ad-free expiration uses server or platform time, not only the device clock.

- Season entitlement and season progression are stored separately.

15.2 Offline Progress

Offline play is a core promise, especially for solo survival and Infinity Taps. Local progress should be versioned and later synchronized with the account. The server should apply reasonable validation limits to coin deltas, survival times, tap rates, and impossible event combinations without punishing legitimate long offline sessions.

15.3 Anti-Cheat Focus

- Impossible tap-rate detection and multitouch filtering.
- Invalid survival times or physics states.
- Duplicate reward transaction replay.
- Forged ownership or loadouts.
- Client-submitted ranked wins without room confirmation.
- Device-clock manipulation affecting streaks, seasons, or ad-free access.
- Automated Infinity Taps farming patterns, while preserving accessibility and legitimate fidget play.

15.4 Analytics Events

Area	Events
Acquisition / onboarding	install, first launch, account created, first mode selected, tutorial completed.
Core loop	round started, round ended, survival time, retry delay, gravity reached, pickup counts.
Retention	sessions, days active, daily/weekly challenge completion, streaks.
Infinity Taps	valid taps, session duration, zones reached, interactions, rare discoveries, exclusive unlocks.
Multiplayer	queue time, bot/human match, disconnects, rematches, rating changes.
Economy	coins earned, spent, purchased, balance, cosmetic purchases, shop opens.
Monetization	ad impressions, ad-free purchase, season pass, supporter bundle, coin packages.
Content	character use, skin/effect equip, badge display, biome visits.

15.5 Validation Metrics

Before investing heavily in real-time multiplayer scale or a large cosmetic catalog, TAM should prove that players voluntarily restart and return because the game feels good. Internal test targets may include strong first-session completion, multiple retries, meaningful Day-1 and Day-7 retention, achievement engagement, profile customization, shop interest, and optional ad participation. These are decision thresholds, not promises.

16. MVP, Development Phases, and Roadmap

16.1 Product Scope Principle

The project should avoid building the entire networked, seasonal, exploratory universe before proving the tap. Competitive TAM and Infinity Taps should be developed as focused slices that share infrastructure without forcing either mode to carry the full complexity of the other.

16.2 Phase 0 — Core Feel Prototype

- One character and one approved six-frame jump sheet.
- Tap impulse, momentum, gravity escalation, ground loss, and rapid restart.
- Green Apple, Poison Fruit, and Golden Hero Token prototypes.
- Basic audio, particles, haptics, and phone deployment.
- Tune input latency and close-call feel before expanding systems.

16.3 Phase 1 — Competitive MVP

- Offline solo survival.
- Personal records and round statistics.
- Four to six characters or skins.
- Small shop and cosmetic loadout.
- Approximately 12–15 achievements and basic badges.
- One primary badge and three showcase badges.
- Daily challenge foundation.
- Analytics and crash reporting.

16.4 Phase 1B — Infinity Taps MVP

- One small natural biome.
- No loss or failure.
- Basic horizontal and vertical exploration.
- Reactive water, bush, tree, and one or two animals.
- Lifetime tap tracking and 1 Hero Coin per 1,000 valid taps.
- One hidden cosmetic or character variant that cannot be purchased.
- Simple discovery journal entry flow.

16.5 Phase 2 — Accounts and Online Foundations

- Email signup, confirmation, login, refresh, and profile provisioning.
- Google login on Android.
- Cloud-synchronized profiles, ownership, achievements, and badges.
- Private online two-device matches.
- Disconnect and result handling.
- Production reward and loadout validation.

16.6 Phase 3 — Public Multiplayer and Bots

- Casual matchmaking and lobby UX.
- Bot backfill with human-like imperfections and visible identification.
- Survival-time matchmaking bands.
- Glicko-2 ranked foundation and Bronze through Grandmaster tiers.
- 90-day seasonal resets and ranked rewards.

16.7 Phase 4 — Biome Journey and Live Content

- Two-minute Infinity Taps zones and branching routes.

- Biome-specific musical layers and ambience.
- More reactive animals, objects, and hero encounters.
- Rare discoveries with pity support.
- Seasonal worlds and environmental storytelling.
- Full achievement catalog and tiered badges.
- Season Pass, Supporter Bundle, and mature cosmetic catalog.

16.8 Long-Term Vision

- Multiple planets, deep-space regions, and uncommon alien encounters.
- Tournament and event structures.
- Ghost races and asynchronous friend challenges.
- Expanded character roster and mastery collections.
- More complex environmental interactions and secrets.
- Seasonal arcs that add new biomes, creatures, characters, and mechanics.

17. Risks, Validation, and Success Criteria

17.1 Primary Product Risks

Risk	Why It Matters	Mitigation
Core loop becomes repetitive	Achievements cannot rescue tapping that is not satisfying.	Prioritize feel, close calls, gravity curve, variation, and rapid testing.
Multiplayer cold start	Empty queues make a functioning game feel dead.	Bot backfill, private matches, solo, and ghost competition.
Scope split	Competitive TAM and Infinity Taps can each become a full game.	Build focused MVP slices and share only systems that truly overlap.
P2W perception	Even small gameplay advantages can undermine ranked trust.	Strict cosmetic-only monetization policy.
Effects harm readability	Paid cosmetics could obscure skill and pickups.	Density caps, reduced-effects mode, shared collision and readability rules.
Infinity tap farming	Automation could distort coin economy and exclusivity.	Valid-tap rules, capped reward cadence, pattern detection, server validation.
Season fatigue	Too much limited content creates pressure and devalues rewards.	Ninety-day cadence, clear progress, restrained catalogs.
Asset inconsistency	Variant sheets may drift in frame count, silhouette, or art quality.	Fixed six-frame template, palette review, nearest-neighbor production checks.

17.2 Core Validation Questions

1. Do players voluntarily restart after losing without being pushed by rewards?
2. Can new players understand the game within seconds?
3. Can skilled players visibly outperform beginners through control rather than raw tapping speed?
4. Does increasing gravity feel fair and readable?
5. Do Green Apples, Poison Fruits, and Hero Tokens create real decisions?
6. Does Infinity Taps feel soothing and alive rather than empty?

7. Do players care about badges and equip them on their profile?
8. Do cosmetic previews make players want to collect without compromising fairness?
9. Can bots fill queues without feeling perfect, deceptive, or insulting?

17.3 Success Definition

Product Success

TAM succeeds when the tap itself is satisfying, players return to improve or explore, cosmetic identity feels meaningful, competition remains fair, Infinity Taps rewards curiosity, and the project can expand seasonally without requiring pay-to-win or unsustainable multiplayer population.

18. Developer Acceptance Criteria

18.1 Core Competitive Build

- Player falls, taps apply responsive upward force, gravity increases, and ground contact ends the round.
- Apple, Poison Fruit, and Hero Token effects are deterministic and recorded in round stats.
- Round result and retry work without network dependency.
- All character skins share the same gameplay collision footprint.
- Tap input, sound, haptic, and animation feedback are measurable on a physical Android phone.

18.2 Infinity Taps Build

- No loss condition or ground-fail state.
- Tap-location movement supports exploration.
- Valid taps update session, lifetime, and biome counters.
- Every 1,000 valid taps grants exactly one Hero Coin once.
- At least one water, bush, tree, and animal interaction has a repeated-tap escalation.
- An Infinity-exclusive reward can be earned but cannot be bought or equipped by an unqualified profile.

18.3 Achievements and Badges

- Progress increments once per event and survives reload.
- Reward transactions cannot duplicate.
- Secret requirements remain hidden until completion.
- Coin-balance achievements evaluate before adding their own reward.
- Players cannot equip unowned badges.
- One primary and no more than three showcase badges are enforced.
- Profile, match results, and leaderboard views render equipped badges consistently.

18.4 Commerce

- Coin packages grant the correct amount after platform validation.
- The one-time starter pack cannot be purchased repeatedly.
- Ad-free access expires after 30 or 60 days as appropriate.
- Season Pass ownership does not automatically grant unearned tiers.
- Earned seasonal cosmetics remain after expiration.
- No product definition contains competitive power.

18.5 Online and Ranked

- Two separate accounts on two phones can authenticate, enter a room, count down, play, and receive a finalized result.
- Disconnects do not duplicate rewards or corrupt rating.

- Bots obey the same physics and show humanized response delay.
- Glicko-2 updates and survival bands are logged and testable.
- Season reset changes ranked state without removing permanent ownership.

Appendix A — Canonical Decision Register

Decision	Canonical Value
Primary platform	Android-first, portrait, Godot.
Core loss condition	Touching ground ends the competitive round.
Difficulty escalation	Gravity increases as the round continues.
Helpful fruit	Green Apple increases jump force.
Debuff fruit	Purple Poison Fruit increases gravity.
Currency pickup	Golden Hero Token; wallet currency is Hero Coins.
Infinity loss	None.
Infinity coin rate	1 Hero Coin per 1,000 valid taps; coins do not regularly spawn.
Infinity exclusives	Cannot be purchased; must be earned in Infinity Taps.
Rank system	Glicko-2 with tiers and survival-time-informed matchmaking.
Season duration	90 days.
Badge loadout	One primary + up to three showcase badges.
P2W	Not allowed.
Ad-free	\$2.99 for 30 days.
Season Pass	\$4.99.
Supporter Bundle	\$9.99 with character bundle, particle, tap effect, and 60 days ad-free.
Slogan	Every Tap Against Me Takes You Further.

Appendix B — Open Balancing Variables

- Exact gravity curve and maximum gravity behavior.
- Base tap impulse, momentum retention, and boundary behavior.
- Apple jump-force magnitude and duration/stacking rules.
- Poison Fruit gravity magnitude and stacking rules.
- Hero Token spawn rate and token-to-coin conversion.
- Competitive daily Hero Coin earning rate.
- Close-call distance and near-ground threshold.
- Ad cadence and rewarded-ad values.

- Ranked queue widening, bot usage thresholds, and survival bands.
- Rare discovery probability, interval, and pity counter.
- Season progression length and challenge point values.

Appendix C — Glossary

Term	Definition
TAM	Tap Against Me.
Competitive TAM	Outcome-driven survival modes with ground elimination, rankings, and opponents.
Infinity Taps	No-loss exploration/fidget mode with persistent taps and exclusive rewards.
Hero Coin	Persistent wallet currency used for cosmetic purchases.
Golden Hero Token	Physical competitive-game pickup that contributes to Hero Coin earnings.
Valid tap	A gameplay tap accepted for movement or reward tracking after input validation.
Primary badge	The badge displayed beside a player's username.
Showcase badge	One of up to three additional badges displayed on the profile.
Season	A 90-day ranked and content period.
Bot backfill	AI opponents added when human matchmaking population is insufficient.
Ghost	A replay of validated human input or trajectory used for asynchronous competition.
Pity system	A mechanism that increases or guarantees a rare reward after repeated unsuccessful opportunities.

Appendix D — Immediate Next Development Actions

1. Lock the Classic Velvet Hex six-frame production template and create a corrected Frozen Hex using the same frame structure.
2. Tune the competitive tap, gravity, Apple, Poison Fruit, Hero Token, ground collision, and retry loop on a physical Android phone.
3. Complete Godot-to-Supabase authentication and Google login.
4. Validate two-device private online play through the production Worker and Durable Objects.
5. Humanize bot item response and survival behavior; implement rating plus survival-band matchmaking.
6. Build the achievement event bus and the first 12–15 achievements with badge loadout.
7. Build one reactive Infinity Taps biome with one exclusive unpurchasable reward.
8. Integrate Opus-compressed music and verify the shipped audio budget on Android.
9. Add analytics before broad testing so retention and replay behavior can guide scope.

TAP AGAINST ME

Every Tap Against Me Takes You Further.

End of Game Design Bible — Version 1.0